

VERTEX INSTITUTE OF TECHNOLOGY | Department of Applied Physics

Laser Diode Dynamics, Interferometry & TIA Circuit Record

Course Code: **PHYS-402** | Module: Electro-Optic Measurement Systems

EXPERIMENT RECORD

EXP-2026-04

Status: **VERIFIED & PASSED**

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Date: 2026-07-24
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Station: Bench B-07
Laser Class: Class 3B
(650nm)

Ambient Temp: 21.4 °C
Rel. Humidity: 44 %

I. Executive Summary & Experimental Objectives

This laboratory investigation presents a complete electro-optical characterization of a Distributed Feedback (DFB) Semiconductor Laser Diode ($\lambda_0 = 650$ nm), an ultra-sensitive Michelson Interferometric Displacement System, and a wideband Photodiode Transimpedance Amplifier (TIA) signal conditioning stage.

Key Experimental Objectives

- Measure $I - I - V$ characteristic curves to determine threshold current I_{th} and slope efficiency η_d .
- Quantify fringe visibility V as a function of optical path difference ΔL in a Michelson setup.
- Optimize a high-speed TIA circuit for gain-bandwidth ($f_{-3dB} \geq 12$ MHz) with phase margin $\phi_m \geq 60^\circ$.

Calibrated Test Instruments

Device	Model	S/N	Uncert.
Power Meter	ThorLabs	OP-9842	$\pm 0.8\%$
Oscilloscope	Keysight	MY5412	$\pm 0.5\%$
Piezo Ctrl	ThorPZT	PZ-4410	± 1.2 nm
SourceMeter	Keithley	SM-8821	$\pm 0.05\%$

Safety Precautions & ESD Operating Rules:

- Class 3B laser radiation hazard ($P_{max} = 15$ mW at 650 nm). Alignment must be performed strictly using certified OD4+ laser safety eyewear.
- PIN photodiode and OPA847 op-amp inputs are extremely sensitive to electrostatic discharge ($V_{ESD,max} = 250$ V). Wrist grounding straps were worn throughout all circuit assemblies.

II. Theoretical Foundations & Governing Equations

The optical output power P_{opt} of the semiconductor laser diode above threshold current I_{th} is governed by the differential quantum efficiency η_d :

$$P_{opt}(I) = \eta_d \frac{\hbar\omega}{q} (I - I_{th}) \quad \text{for } I > I_{th} \quad (1)$$

where $\hbar\omega = \frac{hc}{\lambda_0}$ represents the photon energy and $q = 1.602 \times 10^{-19}$ C is the elementary electronic charge.

In the Michelson interferometer configuration, the superposition of the reference beam E_1 and signal beam E_2 yields a total detected optical intensity $I_{det}(\delta)$:

$$I_{det}(\delta) = I_1 + I_2 + 2\sqrt{I_1 I_2} \gamma(\tau) \cos\left(\frac{4\pi\Delta d}{\lambda_0} + \phi_0\right) \quad (2)$$

where Δd is the mirror displacement, $\gamma(\tau)$ is the complex degree of temporal coherence, and the interference fringe visibility V is defined as:

$$V = \frac{I_{max} - I_{min}}{I_{max} + I_{min}} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2} |\gamma(\tau)| \quad (3)$$

For the front-end optical detection circuit, the photocurrent $I_{pd} = \mathcal{R} \cdot P_{opt}$ (where $\mathcal{R} = 0.45$ A/W is photodiode responsivity) is converted to output voltage V_{out} via a transimpedance amplifier. The -3 dB bandwidth f_{-3dB} and optimal feedback capacitance C_f to prevent peaking are given by Butterworth filter criteria:

$$C_f = \sqrt{\frac{C_{in}}{2\pi R_f f_T}}, \quad f_{-3dB} = \sqrt{\frac{f_T}{2\pi R_f C_{in}}} \quad (4)$$

where $C_{in} = C_{pd} + C_{diff} + C_{cm}$ is the total input capacitance and f_T is the Gain-Bandwidth Product (GBWP) of the operational amplifier.